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A Systematic Review on Code Clone Detection

QURAT UL AIN, WASI HAIDER BUTT, MUHAMMAD WASEEM ANWAR, FAROOQUE AZAM, AND BILAL MAQBOOL

Department of Computer and Software Engineering, College of Electrical and Mechanical Engineering, National University of Sciences and Technology (NUST), Islamabad 44000, Pakistan

Corresponding author: Muhammad Waseem Anwar (waseemanwar@ceme.nust.edu.pk)

ABSTRACT Code cloning refers to the duplication of source code. It is the most common way of reusing source code in software development. If a bug is identified in one segment of code, all the similar segments need to be checked for the same bug. Consequently, this cloning process may lead to bug propagation that significantly affects the maintenance cost. By considering this problem, code clone detection (CCD) appears as an active area of research. Consequently, there is a strong need to investigate the latest techniques, trends, and tools in the domain of CCD. Therefore, in this paper, we comprehensively inspect the latest tools and techniques utilized for the detection of code clones. Particularly, a systematic literature review (SLR) is performed to select and investigate 54 studies pertaining to CCD. Consequently, six categories are defined to incorporate the selected studies as per relevance, i.e., textual approaches (12), lexical approaches (8), tree-based approaches (3), metric-based approaches (7), semantic approaches (7), and hybrid approaches (17). We identified and analyzed 26 CCD tools, i.e., 13 existing and 13 proposed/developed. Moreover, 62 open-source subject systems whose source code is utilized for the CCD are presented. It is concluded that there exist several studies to detect type1, type2, type3, and type4 clones individually. However, there is a need to develop novel approaches with complete tool support in order to detect all four types of clones collectively. Furthermore, it is also required to introduce more approaches to simplify the development of a program dependency graph (PDG) while dealing with the detection of the type4 clones.

INDEX TERMS CCD, SLR, code clone detection, CCD tools, code clone types.

I. INTRODUCTION

Code duplication by copying and pasting without or minor modification into another section of code frequently occurs in software development. This copied code is called code clone and the process is called code cloning. If an error is identified in one part of the code, correction is required in all the replicated segments. Therefore, it is essential to identify all related segments throughout the source code. Various studies suggested that almost 20-50 percent of large software systems consist of cloned code [1], [2].

There is no appropriate definition of code clone. Different researchers used different terms for cloning. Krinke [3] utilized the term "similar code". Baxter *et al.* [4] suggested that a clone is "a code segment that is identical to another segment". Ducasse *et al.* [2] utilized the term "duplicated code". Komondoor and Horwitz [5] also used "duplicated code" and clone as an item of duplicated code. Basic types of clones are listed below [6]:

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Exact clones (Type 1): Identical code segments except for changes in comments, layouts and whitespaces.

Renamed clones (Type 2): Code segments which are syntactically or structurally similar other than changes in comments, identifiers, types, literals, and layouts. These clones are also called parameterized clones.

Near Miss clones (Type 3): Copied pieces with further modification such as addition or removal of statements and changes in whitespaces, identifiers, layouts, comments, and types but outcomes are similar. These clones are also known as gapped clones.

Semantic clones (Type 4): More than one code segments that are functionally similar but implemented by different syntactic variants.

Although there are four types of clones, sometimes people use different terms when referring to the clone relation to their experiments. Common terms utilized by them are given below.

Structural clones: Simple clones that follow the syntactic structure of a particular language within the syntactic

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